

30.3 VIP-Sat: A Boolean Satisfiability Solver Featuring 5×12 Variable In-Memory Processing Elements with 98% Solvability for 50-Variables 218-Clauses 3-SAT Problems

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Boolean satisfiability (SAT), a non-deterministic polynomial (NP)-complete problem, has gained increasing attention with applications in artificial intelligence, machine learning, electronic design automation, and VLSI tests with practical examples, such as AI planning and ML model verification, fault diagnosis, combinational equivalence checking and automatic test pattern generation. The SAT problem is based on testing if a Boolean formula in conjunctive normal form (CNF) can be solved by finding a set of Boolean variables that satisfy the formula (i.e., makes the formula TRUE). Conventionally, software-based SAT-solving algorithms, such as WalkSAT, have been widely used. However, they have limitations in solving large-scale SAT problems with an increasing number of variables due to limited parallelism and a significant latency and energy overhead arising from frequent memory access. Alternatively, dedicated ASIC hardware SAT solvers have demonstrated solving SAT problems using various methods, such as in-memory computing [1], continuous-time dynamics [3], recurrent neural networks [4], and digital processing elements [6]. While most prior works [3– 6] have demonstrated low solvability for hard problems with a large number of variables, a recent work [1] has achieved 72% solvability for solving a hard 3-SAT problem (i.e., a reduced SAT problem with at most 3 literals per clause) with 60 variables in 0.713ms while consuming 1.098μJ, which outperforms prior hardware solvers and achieved ~10× speed-up compared to a CPU (Xeon W-2195 at 4.2GHz). However, the prior work [1] operates based on a popular SAT-solving algorithm (WalkSAT), which may limit further advancement in solvability, solution time and energy consumption. In this work, we propose a novel algorithm-hardware co-designed SAT solver with a digital in-memory processing element architecture incorporating a key feature of parallel uncorrelated variable processing element (VPE) for column-by-column fast SAT solving operations



with high solvability and minimum energy consumption. Measured results show 98% solvability for 50 variables and 218 clauses in 18.7 μ s consuming 20.8nJ.

[Figure 30.3.1](#) overviews the proposed SAT solver (VIP-Sat) with key features and design highlights. In contrast to the prior work [\[1\]](#) and other software algorithms, which sequentially update a randomly picked single variable, the proposed solver can speed up SAT-solving operations by processing multiple uncorrelated variables (i.e., variables with no shared clause) in the same column of the proposed VPE array. As a result, we achieved a solution time of less than 100 column update clock cycles for solving 3-SAT problems with 50 variables, while the prior work [\[1\]](#) and software algorithms require $\sim 10,000$ clock cycles of sequential operations, as shown in the flow chart. Besides the high solvability with reduced solution time, the proposed in-memory architecture helps minimize energy consumption by directly reading memory with no write operations for updating variables using traditional SRAM I/O periphery. The implemented 5×12 VPE array (for solving up to 50 variables) can be scaled up to a 10×12 array to solve 3-SAT problems with 100 variables, demonstrating the scalability of the proposed architecture.

The overall architecture with key circuit building blocks is illustrated in [Fig. 30.3.2](#). A 65nm test chip prototype implemented a 5×12 VPE array, where each column comprises five VPEs operating on the same clock. A VPE comprises three parts, including 32 clause columns, an adder tree, and a variable update logic with a variable register. A register located in the top-right corner of the VPE circuit broadcasts the updated variable through variable buses horizontally, connecting all the VPEs in the same row. The updated 60 variables are broadcast to the buses, and each VPE can select up to 32 correlated variables (i.e., variables with shared clauses) and route them to clause logic located at the top of the clause column using a pair of transmission gates and two 6T SRAM bitcells storing the variable presence information of each clause. An operation logic and circuit diagram of the clause logic is shown in [Fig. 30.3.2](#) (bottom-left), where it generates -1 ($c_i c_o = 11$) or $+1$ ($c_i c_o = 01$) if the clause assigned to the column is unsatisfied with the selected two variables, while the sign of the VPE's variable is 0 ($\neg V_i$ or an inverted variable) or 1 (V_i or a non-inverted variable). The 32 parallel outputs from the clause columns are summed up using an adder tree between the clause columns and the variable update logic. The adder output is connected to the input of the variable update logic, as shown in [Fig. 30.3.2](#) (bottom-right). The VPE's variable is updated to 1 (TRUE) if the adder output is positive (the MSB of the adder output is 0), indicating that there are more unsatisfied clauses with the VPE's variable non-inverted. If the adder output is negative (the MSB of the adder output is 1), the VPE's variable is updated to 0 (FALSE), which indicates that there are more unsatisfied clauses with the VPE's variable inverted. Note that the logic inverts the variable when the adder output is zero, where there are two possibilities (one is when there is no unsatisfied clause, and the other is when there is a balanced number of unsatisfied clauses with the VPE's variable inverted and non-inverted). Our decision to invert the variable was based on behavioral simulation results showing higher solvability for the inverting case.

[Figure 30.3.3](#) (left) shows the proposed SAT-solving workflow. Before the SAT-solving operation (initialize random variables, 12 parallel processing steps of columns of VPEs, and iterations in case the SAT problem is unsatisfied), we first map all the variables to the VPE array based on a mapping rule (i.e., place uncorrelated variables in the same column for maximum parallelism), as shown in [Fig. 30.3.3](#) (top-right). A 50-variable correlation map, which corresponds to a 3-SAT problem #3 with 50 variables and 218 clauses (uf50-218) from a well-known SAT benchmark library, SATLIB [\[2\]](#), shows how variables are correlated with each other, where white (or black) pixels indicate correlated (or uncorrelated) variable pairs. Based on the variable mapping rule, we mapped all 50 variables to 12 columns of 5 VPEs, enabling 5 parallel variable updates, while there are 10 unused VPEs (83.3% utilization). Note that the proposed SAT-solving algorithm combined with parallel uncorrelated VPE processing results in an overall solution time speed-up of $38\times$ with $53\times$ lower energy consumption compared to the prior best solution [\[1\]](#), while achieving 98% solvability for 50 variables. For the column-by-column VPE processing, we use 12-phase low-frequency clocks, as shown in [Fig. 30.3.3](#) (center-bottom). The simulated transient behavior in [Fig. 30.3.3](#) (bottom-right) shows a quickly decreasing number of unsatisfied clauses in less than 100 column update clock cycles for 3-SAT problems with 50 variables (mean solution time for 20/50 variables is 45.6/94.4 cycles).

[Figure 30.3.4](#) illustrates a critical path of the VPE array, from VPE #59 at the bottom of the last column to VPE #0 at the top of the first column. We added an extra clock delay (t_2) for the critical path, which is longer than the nominal delay (t_1) between the adjacent columns to maximize the operating speed (or minimize the solution time), while minimizing accuracy degradations due to bit errors caused by the critical path. [Figure 30.3.4](#) (bottom) shows the measured accuracy when the 50-variable update time ($T_{V50} = 11 \cdot t_1 + t_2$) is swept from 150ns to 220ns. 100% accuracy is achieved when the 50-variable update time is 220ns, resulting in minimal solution time with 100% solution accuracy and no bit errors.

[Figure 30.3.5](#) shows measurement results with two sets of 3-SAT benchmark problems with different numbers of variables (20/50) and clauses (91/218) from SATLIB [\[2\]](#). Note that the ratio of clauses and variables (CTV) indicates the difficulty of the problem, where the 3-SAT problem with a CTV ratio above 4.3 typically indicates a ‘hard’ problem vs. problems with a lower CTV ratio. Both benchmark problems with 20 and 50 variables used in this work have their CTV ratios of >4.3 (i.e., 4.55 for 20 variables and 4.36 for 50 variables). The measured solvability for 20 variables and 91 clauses is 100% at 7 μ s consuming 2.1nJ at 1.2V with the 20-variable update time of 120ns. We achieved 98% solvability for 50 variables and 218 clauses at 18.7 μ s and consumed 20.8nJ at 1.2V. Note that the solution time and energy consumption can be reduced to 1 μ s/0.3nJ (for 20 variables) and 5.1 μ s/5.7nJ (for 50 variables) when the solvabilities are relaxed to 80% (for 20 variables) and 68% (for 50 variables). The number of transient variable transitions dominates the energy consumption, while they are minimal as most transitions happen at the initial operation cycles, as shown in [Fig. 30.3.5](#) (bottom-left). The measured energy consumption for a single iteration is 0.15nJ and 1.89nJ for 20 and 50 variables, respectively, at 1.2V. [Figure 30.3.5](#) (right) shows plots (solvability, solution time, and energy) comparing performance with prior work. [Figure 30.3.6](#) is a table comparing the proposed SAT solver to state-of-the-art works. [Figure 30.3.7](#) shows the 65nm test chip die micrograph and a summary table. The test chip implemented 60 VPEs in the core area of 1.116mm².



Figure 30.3.1:

Introduction to Boolean satisfiability (SAT) problems, applications, and the proposed SAT solver (VIP-Sat) with key features and design highlights.



Figure 30.3.2:

Proposed VIP-Sat overall architecture and key circuit building blocks (VPE, clause logic and variable update logic).



Figure 30.3.3:

Proposed SAT solving workflow and VPE processing mechanism (left); problem mapping, operation timing diagram, and simulated results of 3-SAT solving problems with 20 and 50 variables (right).



Figure 30.3.4:

A VPE's critical path of the column-by-column variable update operations for 3-SAT problems with 50 variables and 218 clauses (top) and measured accuracy results vs. a 50-variable update time (bottom).



Figure 30.3.5:

Measured result for 3-SAT problems with 20 variables/91 clauses and 50 variables/218 clauses and comparisons with prior works in solvability, solution time and energy consumption.



Figure 30.3.6:

Comparison with state-of-the-art hardware SAT solvers.



Figure 30.3.7:
A 65nm VIP-Sat test chip die micrograph and a summary table.

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